

**Comparing Mutual Information in Finite Samples:
Wald and Expanded Welch–Satterthwaite
Inference**

Working Draft

Chapters 1–4, with titles for the remaining chapters

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Contents

1	Introduction	1
1.1	Why the comparison is difficult	1
1.2	Research questions	2
1.3	Contributions to the field	2
1.4	Scope and organisation	3
2	Background and Literature Review	4
2.1	Information-theoretic foundations	4
2.1.1	Entropy and information content	4
2.1.2	Pointwise and average mutual information	5
2.2	Estimating discrete mutual information	6
2.3	Sampling variance of plug-in MI	7
2.4	Approaches to Testing MI Differences	12
2.5	Previous work and this thesis’s contribution	15
3	The Two-Sample MI Test and Wald Baseline	16
3.1	Problem setup and bias-corrected estimator	16
3.2	Shared standard error and test statistic	18
3.3	Normal Wald reference and motivation for expansion	18
4	The Expanded Welch–Satterthwaite Test for MI	20
4.1	Choosing Degrees of Freedom for Expanded Welch	20
4.2	How the MI Variance Changes with the Table	21
4.3	Component degrees of freedom and final test	24
4.4	Worked example and implementation	27
4.5	Behaviour and limitations	28

5	Experimental Design	30
6	Experimental Results	31
7	Discussion	32
8	Conclusion	33

Chapter 1

Introduction

Suppose two independent groups are observed on the same categorical variables X and Y . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the population MI values differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether X and Y are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where P and Q are the two joint distributions. The samples from P and Q are independent of each other; that sampling assumption does not imply independence of X and Y within either population.

1.1 Why the comparison is difficult

MI uses the probabilities in every cell of a table and the totals of its rows and columns. In a small sample, some cells can be empty even when their population probabilities are positive. The observed tables determine both the estimated MI difference and its standard error, so sparse data can affect both. The Normal Wald test compares their ratio with a normal curve, but in small samples it may reject equal MI more or less often than the intended 5%.

Expanded Welch offers a possible adjustment. It uses the same estimated difference and standard error as Wald, but compares their ratio with a Student t

distribution. Its degrees of freedom depend on how much the estimated MI variance changes from sample to sample. To calculate that change, we must account for the fact that changing one cell also changes its row and column totals and the MI values calculated from them. This thesis derives that calculation and tests whether the adjustment helps.

The adjustment is not automatically better. It can reduce false positives where Wald rejects too often, but it can also reduce detection where Wald already rejects too rarely. In sparse samples, it may fail to return a usable p-value when Wald succeeds. The experiment therefore examines false positives, detection and the frequency of usable results together.

1.2 Research questions

The thesis asks two main questions. First, how can changes in the estimated MI variance be used to choose the degrees of freedom for a Student test? Second, does the resulting Expanded Welch test offer a useful improvement over Normal Wald? Answering the second question requires looking at false positives, detection when MI differs, and the frequency of usable results. The experiment checks how these outcomes change with table size, uneven cell probabilities, the amount and location of association, unequal sample sizes and larger samples. It also measures the extra computation required.

1.3 Contributions to the field

This thesis combines established ideas about MI and Welch–Satterthwaite tests for a specific question: whether two populations have equal MI. Its contributions are the calculation needed for that test and evidence about whether the test is useful.

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance. The calculation includes the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this calculation to choose the degrees of freedom for a two-sample Student test. The main experiment compares it with Wald; a supplementary analysis checks simpler sample-size choices for the degrees of freedom.
3. **Empirical contribution.** A reproducible experiment measures false positives, detection, usable results and runtime on fixed populations with known MI. It

covers 2×2 through 8×8 square tables, rectangular tables, unequal samples and very sparse settings. The companion atlas retains the result for each setting rather than hiding it in an overall average.

4. **Practical contribution.** Expanded Welch can help when Wald rejects equal MI too often. It can also make an already-low rejection rate lower and return fewer usable results in sparse samples. The evidence shows why it should not replace Wald in general and points to problems that changing the reference distribution alone cannot solve.

1.4 Scope and organisation

The analysis concerns independent samples recorded in count tables with the same row and column categories. The main derivation assumes every population cell has positive probability and that its leading large-sample MI variance is greater than zero. Exact independence is considered separately because that variance becomes zero. The experiment compares Normal Wald with Expanded Welch using a two-sided 5% test. It does not establish performance for paired data, continuous MI estimators or a number of categories that grows with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 shows the complete landscape before the focused comparisons. Chapters 7 and 8 discuss the consequences and the resulting recommendation. The appendices provide the detailed algebra, construction and source-to-result record.

Chapter 2

Background and Literature Review

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = - \sum_x p_X(x) \log p_X(x). \quad (2.1)$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y) , the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X | Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y .

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \dots, r\}$ and $Y \in \{1, \dots, c\}$. Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.2)$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.3)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.4)$$

Here the summations are nested: for each row i , the inner sum runs over all columns j . Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$. Because P is the joint distribution of (X, Y) , the same quantity is conventionally written as $I(X; Y)$. Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.5)$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells.

MI measures the overall departure from independence, not one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. Our null requires only this one number to agree, not that the two tables match cell by cell.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij} . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.6)$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.7)$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts are random but constrained to sum to n , so they are dependent. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample behaviour depends on the complete probability table. In particular, rare cells can change the sampling behaviour even when the alphabet and sample sizes are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the plug-in estimator has a positive leading bias as the sample size grows:

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.8)$$

The leading term grows with table size and shrinks as the sample grows. It does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.9)$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

2.3 Sampling variance of plug-in MI

To compare two estimated MI values, it is necessary to know how much each estimate would vary across repeated samples. The goal of this section is to derive the large-sample approximation

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.10)$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (2.11)$$

Let $f(u) = u \log u$. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du} \{u \log u\} = \log u + u \frac{1}{u} = \log u + 1. \quad (2.12)$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \quad (2.13)$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \hat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \hat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\hat{u} \log \hat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\hat{u} - u)}_{\text{first-order change}}. \quad (2.14)$$

Subtracting the population value from both sides makes the change explicit:

$$\hat{u} \log \hat{u} - u \log u \approx (\log u + 1)(\hat{u} - u). \quad (2.15)$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} (\log p_{ij} + 1)(\hat{p}_{ij} - p_{ij}) \\ &\quad - \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) \\ &\quad - \sum_j (\log p_{+j} + 1)(\hat{p}_{+j} - p_{+j}). \end{aligned} \quad (2.16)$$

The margin errors are sums of the corresponding cell errors:

$$\hat{p}_{i+} - p_{i+} = \sum_j (\hat{p}_{ij} - p_{ij}), \quad \hat{p}_{+j} - p_{+j} = \sum_i (\hat{p}_{ij} - p_{ij}). \quad (2.17)$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) &= \sum_i \left[(\log p_{i+} + 1) \left\{ \sum_j (\hat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.18)$$

The inner sum over j is the probability error for a fixed row i . It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way. Consequently, the same cell error $\hat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &= \log \left(\frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &= \ell_P(i, j) - 1. \end{aligned} \quad (2.19)$$

Therefore,

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i, j) - 1\}(\hat{p}_{ij} - p_{ij}). \quad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\hat{p}_{ij} - p_{ij}) = \sum_{i,j} \hat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \quad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}
 \hat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}) \\
 &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}) - \sum_{i,j} (\hat{p}_{ij} - p_{ij}) \\
 &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}). \tag{2.22}
 \end{aligned}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a . An empirical cell probability is the proportion of observations in that cell, so

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}
 \hat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^n \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\
 &= \frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}. \tag{2.25}
 \end{aligned}$$

The first sum inside the brackets selects the single cell containing Z_a , giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$. This equation is the key step. To first order, the error in the plug-in MI estimate behaves like the average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$E_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i,j) - I(P)\}$$

$$= \underbrace{\sum_{i,j} p_{ij} \ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_1 = 0. \quad (2.26)$$

In formal asymptotic terminology, the centred function $\ell_P(i, j) - I(P)$ is the influence function of MI; Equation (2.25) is its first-order linear representation (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \text{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.27)$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \text{Var}\{\hat{I}(P)\} &\approx \text{Var}\left\{\frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2} \sum_{a=1}^n \text{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2} \sum_{a=1}^n V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \quad (2.28)$$

This proves the approximation stated in Equation (2.10). It is the familiar result that the variance of an average of n independent contributions is the variance of one contribution divided by n .

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\hat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \quad (2.29)$$

This does not claim that every finite-sample MI estimate is normally distributed. It is a local, large-sample approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i, j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes.

A finite sample can still have positive MI. Its leading variation comes from the

squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limit (Brillinger, 2004; Marinescu and Balcau, 2025). The normal limit for comparing positive MI values and the chi-squared limit for testing independence therefore apply in different settings.

2.4 Approaches to Testing MI Differences

Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference (Wald, 1943; van der Vaart, 1998). For two independent MI estimates, the two first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.30)$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\hat{I}(P) - \hat{I}(Q)}{\sqrt{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q}} \xrightarrow{d} N(0, 1) \quad \text{under } I(P) = I(Q). \quad (2.31)$$

Here $\hat{V}(P)$ and $\hat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test.

With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses a normal reference; it makes no separate small-sample adjustment for that standard-error variability.

The Welch–Satterthwaite principle.

Welch’s test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means $\bar{X}_P, \bar{X}_Q$ and variances s_P^2, s_Q^2 , the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \quad (2.32)$$

$$\nu = \frac{(s_P^2/n_P + s_Q^2/n_Q)^2}{\frac{(s_P^2/n_P)^2}{n_P - 1} + \frac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \quad (2.33)$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both component values affect ν . Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

Welch compares the statistic with a Student distribution using the calculated ν . At the same significance level, this requires a larger absolute statistic to reject the null than a standard normal reference does. The difference between the Student and normal rejection thresholds becomes smaller as ν increases.

For MI, the estimated variance contributions are $\hat{V}(P)/n_P$ and $\hat{V}(Q)/n_Q$. Their component degrees of freedom need their own calculation because $\hat{V}(P)$ and $\hat{V}(Q)$ are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

Regression studies have also adjusted reference distributions to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). They establish the general strategy, but not whether it helps with MI. For example, Bell and McCaffrey (2002) report regression variance estimates biased downward, while Kauermann and Carroll (2001) examine intervals that cover too rarely. A more cautious reference can help in such settings. Near independence, however, the estimated MI variance can be biased upward. A Student reference may then make an already cautious test reject even less often. Section 6.7 examines this possibility.

Prior MI inference and related comparisons.

MI uncertainty also has a broader literature than its leading normal approximation. Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. Hutter (2002) and Hutter and Zaffalon (2005) derive moments of the posterior distribution of MI under a Dirichlet model, including higher-order variance and shape information. Their posterior uncertainty has a different conditioning from the repeated-sampling distribution studied here. Their results establish that higher-order MI uncertainty is not a new topic, but do not directly validate the two-sample Student reference in this thesis.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student reference, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.34)$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s_{H,P}^2$ and $s_{H,Q}^2$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Applying the same assignment to the two MI variance contributions gives a clear historical baseline. It does not, however, derive how the plug-in MI variance itself fluctuates: MI depends on a joint distribution and both margins, so moving mass into one cell changes pointwise values elsewhere in its row and column. The expanded calculation retains these changes.

The question of comparing MI values also has predecessors. In a segregation application, Mora and Ruiz-Castillo (2009) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature (Mora and Ruiz-Castillo, 2011). They do not derive the variance-sensitivity degrees of freedom used here, but the general aim of comparing MI values is not new. Guha and Chothia (2014) use MI for a different two-sample question: whether two continuous distributions are identical. Their MI is between a pooled observation and its sample label; their null is equality of distributions, not equality of two within-population MI values. For discrete data, Drake and Guha (2014) develop an MI-based k -sample test for the same distributional-equality question.

At exact independence, the first-order sampling variation of MI vanishes and its second-order variation becomes important (Marinescu and Balcau, 2025). The positive-MI comparison developed here is therefore not a replacement for a one-table independence test. Second-order effects can also matter near that boundary.

Resampling alternatives.

A bootstrap would draw many new samples from the observed tables to estimate how much the test statistic varies. This requires repeating the calculation many times (Efron, 1979). A simple permutation test instead combines the two samples and shuffles their group labels. That shuffle is justified when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Shuffling labels could therefore test a stronger claim than this thesis asks about (Chung and Romano, 2013). More specialised permutation methods are possible for some such comparisons (Janssen, 1997; Chung and Romano, 2013), but assessing one for equal MI would be a separate study. The present comparison focuses on Normal Wald and Expanded Welch.

2.5 Previous work and this thesis's contribution

Previous research explains how to estimate MI, correct its leading small-sample bias and approximate its sampling variance. Wald's method uses that variance to compare two estimates. Welch–Satterthwaite methods also allow for uncertainty in an estimated variance. Researchers have also compared entropy and MI-related quantities.

The missing step for this test is to measure how much the *estimated MI variance* changes from sample to sample. Assigning each population degrees of freedom based only on its sample size, as earlier information-based and Welch tests do, does not measure this change. A new observation affects a cell probability, its row and column totals, and the MI values calculated from them. This thesis derives how those changes affect the estimated variance and uses the result to choose a Student t distribution for comparing two MI values. It then tests the method on fixed population tables, showing when it brings false-positive rates closer to the target and when it instead rejects too rarely.

Chapter 3 defines the comparison, Chapter 4 derives the adjustment, and Chapter 6 evaluates it.

Chapter 3

The Two-Sample MI Test and Wald Baseline

This chapter defines the population MI difference being tested and the statistic calculated from the observed tables. Both methods use this same statistic. Normal Wald judges it against a standard normal distribution; Chapter 4 develops a Student distribution with estimated degrees of freedom.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent $r \times c$ count tables with totals n_P and n_Q . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling). *Observations are independent and identically dis-*

tributed within each population, and the two population samples are independent.

Assumption 2 (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities). *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI). *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units). *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives $\hat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\hat{\Delta}_{\text{BC}} = \left\{ \hat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \hat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \quad (3.7)$$

The two terms in braces are the bias-corrected MI estimates for the separate populations.

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. A corrected estimate can be negative in a sparse sample and is not clipped at zero, since clipping would define a different nonlinear estimator. When

$n_P = n_Q$, the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\hat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i,j) - \hat{I}(P)\}^2. \quad (3.8)$$

Define $\hat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.9)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.10)$$

The null hypothesis is not imposed when calculating $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$ or T : they use the observed tables. Equal population MI enters when interpreting the reference distribution, which is centred at zero under H_0 . Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald reference and motivation for expansion

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \quad \text{under } H_0. \quad (3.11)$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.12)$$

where Φ is the standard normal distribution function.

The normal reference is a large-sample approximation. Wald estimates the standard error from the tables but applies no degrees-of-freedom adjustment.

Why consider an expanded reference?

The shared statistic supplies the estimated effect and its first-order standard error. Earlier information-theoretic and ordinary-Welch analogues set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how much the plug-in MI variance changes when the observed table changes. That variance depends on joint cells, margins and pointwise MI values. Expanded Welch instead estimates each component's degrees of freedom from how the complete MI variance estimate changes with the table. The supplementary analysis also compares these sample-size assignments with Expanded Welch.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P , to first order,

$$E\{\hat{V}(P)\} \approx V(P). \quad (3.13)$$

Chapter 4 derives $\text{Var}\{\hat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3. It changes the distribution used to judge that statistic from normal to Student. To choose the Student degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples.

4.1 Choosing Degrees of Freedom for Expanded Welch

For the two MI estimates, the Welch–Satterthwaite equation combines their estimated variance contributions as in the original moment-matching framework (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \nu_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \nu_Q}. \quad (4.1)$$

The numerator contains the two terms in the squared standard error. The remaining quantities are the component degrees of freedom ν_P and ν_Q , which should measure how reliably $\hat{V}(P)$ and $\hat{V}(Q)$ estimate their population counterparts.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\hat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\hat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one component.

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, the working approximation represents $\widehat{V}(P)$ by

$$\mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.2)$$

The approximation matches moments; it does not assert that the finite-sample distribution of $\widehat{V}(P)$ is exactly chi-squared.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathbb{E}(\chi_{\nu_P}^2) = \nu_P, \quad \text{Var}(\chi_{\nu_P}^2) = 2\nu_P. \quad (4.3)$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathbb{E} \left[\mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \mathbb{E}\{\widehat{V}(P)\}. \quad (4.4)$$

Its variance is

$$\text{Var} \left[\mathbb{E}\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P} \right] = \frac{[\mathbb{E}\{\widehat{V}(P)\}]^2}{\nu_P^2} \text{Var}(\chi_{\nu_P}^2) \quad (4.5)$$

$$= \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \quad (4.6)$$

Matching this quantity to $\text{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathbb{E}\{\widehat{V}(P)\}]^2}{\text{Var}\{\widehat{V}(P)\}}}. \quad (4.7)$$

To first order, $\mathbb{E}\{\widehat{V}(P)\} \approx V(P)$. The required calculation is therefore the sampling variance of $\widehat{V}(P)$.

4.2 How the MI Variance Changes with the Table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.8)$$

where $\delta_{(x,y)}$ places probability one on cell (x, y) . At $\varepsilon = 0$, the distribution is P . Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon \mathbf{1}\{i = x, j = y\}. \quad (4.9)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.10)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.11)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.12)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.13)$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.14)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.15)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.16)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{ \mathbf{1}(i=x, j=y) - p_{ij} \} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.17)$$

The first sum is $\ell_P(x, y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1 ; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.18)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.19)$$

Write this variance in second-moment form:

$$V(P) = \text{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \quad (4.20)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.21)$$

Differentiating Equation (4.20) gives

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} \text{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

It remains to differentiate

$$\text{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.23)$$

The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.24)$$

The first sum is

$$\ell_P(x, y)^2 - E_P \{ \ell_P(X, Y)^2 \}. \quad (4.25)$$

For the second sum, define the probability-weighted row and column means

$$E_P \{ \ell_P(X, Y) \mid X = x \} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.26)$$

$$E_P \{ \ell_P(X, Y) \mid Y = y \} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.27)$$

Substitution of Equation (4.15) gives

$$\begin{aligned} &\sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}} - \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}} + I(P) \\ &= \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P). \end{aligned} \quad (4.28)$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) yields

$$\boxed{\begin{aligned} g_P(x, y) &= \{ \ell_P(x, y) - I(P) \}^2 - V(P) \\ &\quad + 2 \left[\ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} \right. \\ &\quad \left. - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P) \right]. \end{aligned}} \quad (4.29)$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for changes in those values, including the effects of the selected joint probability and both margins.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \quad (4.30)$$

Averaging the one-cell perturbation directions gives $\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = 0$. The derivative is linear in its direction, so $E_P\{g_P(X, Y)\} = 0$; Appendix A shows the steps. Explicit centring protects the numerical calculation if roundoff produces a small nonzero empirical mean.

Assumption 6 (Regular variance sensitivity). *For the first-order Expanded Welch construction, $\tau^2(P) > 0$ and $\tau^2(Q) > 0$.*

The empirical distribution is

$$\hat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\delta_{Z_a^{(P)}} - P\}. \quad (4.31)$$

Taylor's first-order term is linear in this difference (van der Vaart, 1998, Chapter 20). By Equation (4.21), each summand has sensitivity $g_P(Z_a^{(P)})$. Therefore,

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

Equation (4.35) depends on relative, not absolute, uncertainty. Since

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.36)$$

the component degrees of freedom are approximately twice the reciprocal of this relative variance. A stable estimate relative to the size of $V(P)$ has large degrees of freedom; a relatively unstable estimate has smaller degrees of freedom and produces a heavier-tailed final reference. This first-order component formula assumes $\tau^2(P) > 0$.

Positive $V(P)$ alone does not guarantee that condition. A small change in the table can have no first-order effect on $V(P)$, even when MI and $V(P)$ are positive. In that case, a higher-order analysis of $\hat{V}(P)$ would be needed to describe its finite-sample distribution.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[\hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.37)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2. \quad (4.38)$$

The estimated component degree of freedom is

$$\boxed{\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}}. \quad (4.39)$$

Repeating the same calculation for the second table gives $\hat{\nu}_Q$.

Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\{\hat{V}(P)/n_P\}^2 / \hat{\nu}_P + \{\hat{V}(Q)/n_Q\}^2 / \hat{\nu}_Q}}. \quad (4.40)$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.41)$$

where F_{t_ν} denotes the cumulative distribution function of a Student t variable with ν degrees of freedom.

The expanded calculation does not change $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$, or T . It changes only the

reference distribution used to evaluate that statistic.

For a fixed finite T , the Student reference gives a two-sided p-value at least as large as the Normal Wald reference. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. Thus every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. The correction cannot raise rejection where Wald is already too conservative. Appendix A proves this ordering and examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.42)$$

Their totals are $n_P = 105$ and $n_Q = 100$. The plug-in MI estimates are 0.13693 and 0.25848 nats. Since $(3 - 1)(3 - 1) = 4$, the corrected difference is

$$\hat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \quad (4.43)$$

The empirical influence variances are $\hat{V}(P) = 0.27189$ and $\hat{V}(Q) = 0.43371$. Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.44)$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the table-based sensitivity calculation first gives $\hat{\tau}^2(P) = 0.93212$ and $\hat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.45)$$

Equation (4.40) combines them into $\hat{\nu}_{\text{expanded}} = 59.03$, giving $p = 0.15264$. Both methods fail to reject at the 5% level for these tables. The example shows that Expanded Welch changes the reference, not the observed statistic.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\hat{V}(P)$, $\hat{V}(Q)$, the shared standard error, and T .
4	Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$.
5	Compute $\hat{\tau}^2(P)$, $\hat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the Student p-value, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$. The method is deterministic once the two count tables are given.

4.5 Behaviour and limitations

Near independence.

The adjustment has a local explanation. Let $P_t = P_0 + tH$, where $P_0 = (q_{ij})$ is independent with positive cells $q_{ij} = a_i b_j$, and H has zero row and column sums. Thus $t = 0$ is the expansion point and the margins stay fixed. For $f(t) = I(P_t)$, the second-order Taylor expansion has $f(0) = 0$, $f'(0) = \sum_{i,j} H_{ij} = 0$, and $f''(0) = \sum_{i,j} H_{ij}^2 / q_{ij}$.

$$I(P_t) = \underbrace{0}_{\text{population value}} + \underbrace{0t}_{\text{first order}} + \underbrace{\frac{t^2}{2} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}}}_{\text{second order}} + O(t^3), \quad (4.46)$$

$$\ell_{P_t}(i, j) = \log \left(1 + \frac{tH_{ij}}{q_{ij}} \right) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \quad (4.47)$$

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.48)$$

Appendix D gives the second-order MI result when the margins can also change. The row and column conditional means of pointwise MI have no first-order term because the corresponding sums of H are zero. Equation (4.29) then gives $g_{P_t}(i, j) =$

$2tH_{ij}/q_{ij} + O(t^2)$, and hence

$$\tau^2(P_t) = 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.49)$$

For a distribution close to independence on this path, these expansions give

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.50)$$

Substitution into the component formula then gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx n_P I(P). \quad (4.51)$$

Thus low MI can produce a small effective degree of freedom and a much heavier-tailed reference. The approximation is local: it assumes fixed table dimension, positive probabilities and sufficiently regular cell counts. Very near independence, quadratic sampling variation can still be important, so a first-order Student adjustment need not improve finite-sample calibration.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) describe how MI and its estimated variance change when the underlying table changes slightly. For tables with positive cell probabilities, these changes give their first-order large-sample approximations.

The scaled chi-squared model in Equation (4.2) and the Student reference are approximations, not exact finite-sample distributions. In particular, $\hat{I}(P)$ and $\hat{V}(P)$ come from the same sample and can vary together. Adjusting the reference for uncertainty in $\hat{V}(P)$ does not account fully for this dependence or make the statistic exactly Student distributed. Appendix A gives the first-order covariance. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student reference approaches the normal reference used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error. Expanded Welch also needs positive, finite degrees of freedom for each population and for their combination. If a requirement fails, the calculation is recorded as invalid, not as evidence for the null. Appendix E gives the exact software rules. Chapters 5 and 6 report how often valid results are obtained and whether the correction improves calibration.

Chapter 5

Experimental Design

Chapter 6

Experimental Results

Chapter 7

Discussion

Chapter 8

Conclusion

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